Instructor Guide: Clean the OrbitCo Incident Export

Type: Hands-on lab (individual work) **Time:** 20 minutes (15 min individual work + 5 min debrief)

Pre-Activity Setup

- Distribute `orbitco_tickets_raw.csv` to every learner (shared drive, LMS download, or USB — whatever the venue supports). Confirm the file opens before class; a file that opens in Numbers on macOS without going through CSV import will quietly lose the leading-zero ticket IDs.
- Have a projected "bad before" screenshot and "clean after" screenshot of the same ticket range ready to show at the debrief. The visual contrast lands harder than a verbal comparison.
- Open a blank workbook on your own screen so you can follow along with the class on any step a learner gets stuck on.
- Ensure every learner has Excel on the desktop (not Excel Online — Text to Columns and Data Validation work differently in the web version and will confuse learners in the 20-minute window).

Facilitation Steps

1. **0:00 — Launch** (2 min): Project the Scenario section and the six known problems list (stray row, duplicates, date formats, blanks, whitespace, case). Ask: "Which of these would you find by eye? Which only shows up when a formula misfires?" Do not answer — the lab answers it. Start the timer.

2. **0:02 — Circulation, first 5 minutes** (focus: steps 1-4): Watch for learners editing `tickets_raw` in place. This is the single most common early failure. If you see it, stop them at the whole class level with: "If you are editing the raw sheet, please stop. That sheet is your evidence — treat it like a crime-scene photo." Continue circulating.

3. **0:07 — Circulation, middle 5 minutes** (focus: steps 5-8): The TRIM/PROPER helper-column sequence trips people up because they forget to paste-as-values before deleting the helper. Watch for the pattern: they compute the helper, delete

it, and the original column still has the raw text. Walk them through paste-values once individually; then project the sequence if more than three learners hit it.

4. **0:12 — Circulation, last 3 minutes** (focus: steps 9-12): The dedupe step is where the "exactly 7" check matters. Any other number means the dedupe key is wrong. Praise the learners who noticed out loud.
5. **0:15 — Stop hands** (debrief, 5 min): Run the three debrief prompts below. Close with the "next lab" hook: the `Tickets` Table they just built is the input to the Key Performance Indicator (KPI) summary lab.

What to Circulate For

- **Editing the raw sheet.** The single most damaging early mistake. Learners who edit `tickets_raw` have broken the audit trail and will not notice. Stop them immediately, have them delete the edited `tickets_raw` and re-import the CSV, and move on. Call it out publicly — this is a lesson worth interrupting for.
- **Deleting missing-severity rows instead of flagging.** The instinct is overwhelming because a spreadsheet with no blanks "looks cleaner." Stop them with: "How many tickets did OrbitCo get in April?" If they answer 394 (413 – 19), ask what happened to the 19. That conversation is the teach.
- **Reformatting dates instead of using Text to Columns.** Superficially the cell now shows a date, but the underlying type is still text. The downstream `ResolutionMinutes` formula produces `#VALUE!`. If a learner hits `#VALUE!`, check the `OpenedAt`/`ClosedAt` columns — they will be left-aligned (text) rather than right-aligned (dates).
- **Deduplicating with all columns selected.** The checkbox default in the Remove Duplicates dialog is "all columns." If a learner clicks OK without unchecking, the reopen-bug duplicates pass through because their timestamps differ slightly. The final row count will be wrong (probably 419 instead of 413). Catch this before they move on.
- **Missing the Table-rename step.** They Ctrl+T, the Table is named `Table1`, they move on. Structured references still work but the formulas read like `=COUNTIF(Table1[Severity], "High")`, which is far less maintainable. Ask them to rename to `Tickets` on the Table Design tab before the lab ends.

Common Pitfalls

- **Row 47 deletion on the wrong sheet.** Some learners delete row 47 on `tickets_raw` by habit before copying to `tickets_clean`. The raw sheet is now corrupted. Recovery: re-import from the CSV.
- **Trailing whitespace on TicketID not propagating.** When they paste-values the `TRIM` helper back, if they paste into only the visible cells instead of the whole column, some rows keep the trailing space. `COUNTIF` on TicketID later reads "T-10047 " and "T-10047" as different values. Workaround: select the entire `TicketID` column when pasting values.
- **Mixed-case severity still mixed after `PROPER`.** `PROPER("HIGH ")` returns `"High "` — the trailing space survives. That is why the helper formula is `=PROPER(TRIM([@Severity]))`, in that order. If a learner wrote `=TRIM(PROPER([@Severity]))` they may be fine; if they wrote only `=PROPER([@Severity])`, the space survives and `COUNTIF` still fails later.
- **Data validation applied before case normalization.** If they apply validation to Severity first, the "HIGH" and "high" values flag as invalid and the dropdown appears to reject them. The right order is clean first, validate second.
- **Save as `.csv` instead of `.xlsx`.** Every formula, Table, and validation rule is lost on save. Excel warns, but the warning dialog is easy to dismiss. Remind them explicitly — "save as `.xlsx`" — at step 12.

Debrief Prompts

- **"What happens if you run Remove Duplicates before trimming whitespace and normalizing case?"** Expected answer: the trailing-space and mixed-case variants get treated as distinct values, so fewer duplicates are detected and the row count does not drop to exactly 413. The order of cleanup steps matters — normalize before dedupe.
- **"Why flag missing-severity rows instead of deleting them? What does your total ticket count read in each case?"** Expected answer: deleting silently changes the total to 394 and leadership will ask where the other 19 went. Flagging preserves 413 total and lets severity-specific KPIs exclude the blanks honestly. This is the defensibility lesson in miniature.

- **"What is the first thing the next person to open this workbook should be told?"** Expected answer: what was done to the raw data, in what order, and why the missing-severity rows are still in the Table. This is the pitch for a `README` tab in every shared workbook — drop the term now and it will pay off in Lab 2 and the dashboard lab in the next lesson.

Expected Answers

The expected end-state of the `Tickets` Table:

Metric	Expected value
Total rows	413
Rows with `SeverityIsMissing = TRUE`	19
Distinct values in `Severity`	High, Medium, Low, (blank)
Duplicates removed at step 10	Exactly 7
`ResolutionMinutes` column type	Numeric, no `#VALUE!` errors on rows with both timestamps
`tickets_raw` sheet	Unmodified from the CSV import (spot-check any row)

If a learner's numbers differ, walk back through the steps in order — mismatches almost always trace to a skipped trim/case-normalize step or a wrong dedupe key.